PRODUCT REQUIREMENT DOCUMENT (PRD)

Project Title: Okeho Ground-Truth Field Capture & Mapping System
Document Version: 1.1 (Performance & Mobile Stability Update)
Target Region: Okeho, Oyo State, Nigeria
Document Owner: Engineering & Operations Lead
Date: August 2026

---

1. Executive Summary & Objectives

1.1 Problem Statement

Inaccurate and missing street network data, spatial boundaries, and Points of Interest (POIs) in Okeho, Oyo State, hinder local navigational accuracy and municipal planning. Standard public edits submitted directly to third-party commercial platforms (such as Google Maps) suffer from extended review delays or rejections.

1.2 System Purpose

Deploy an offline-first mobile data capture application paired with a PostGIS geospatial backend pipeline to collect, validate, and normalize ground-truth mapping data into standard GeoJSON constructs.

1.3 Dual-Distribution Strategy

1. **OpenStreetMap (OSM) Pipeline:** Immediate publishing of street geometry, landmarks, and spatial vectors to OSM via automated/semi-automated API ingest pipelines without third-party gatekeeping.
2. **Google Maps / Government Submissions:** Bulk submission of institutional-grade, verified GeoJSON/KML payloads via official local government channels and Google Geo Partner contacts.

---

2. Core Field Workflows

2.1 Point Feature Capture Workflow (POIs & Landmarks)

Field collectors walk directly to a landmark or structure to record a point feature:

1. **GPS Engine Trigger:** Initiates 3–5 second continuous stationary position sampling.
2. **Metadata Recording:** Records latitude, longitude, elevation, and the hardware accuracy radius (in meters).
3. **Threshold Check:** If accuracy exceeds 15 meters, the system prompts the worker to step into a clearer line-of-sight location.

4. **On-Device Media Capture:** Captures high-resolution images of storefronts, address plates, or street signs.
5. **On-Device OCR Execution:** Runs lightweight text extraction to provide interactive text suggestions.
6. **Data Storage:** Saves the compiled record into the local offline SQLite database queue.

2.2 Line Feature Capture Workflow (Street Segments)

Street segments utilize multi-point coordinate sampling with spatial interpolation for curves and intersections:

1. **Start Segment:** Field worker taps “Start Segment” at an intersection or street boundary (Point A).
2. **Midpoints & Curves:** Field worker taps “Add Control Point” at curves, midpoints, or bends (Points B, C, etc.).
3. **End Segment:** Field worker taps “End Segment” at the terminal node (Point D).
4. **Attribute Entry:** Inputs street name, surface type (paved/unpaved), and traffic directionality (one-way/two-way).

---

3. Technical & System Requirements

3.1 Mobile Capture Application (Flutter Client)

FR-1.1: GPS Location Engine & Stationary Sampling

- **Hardware Utilization:** Uses internal GNSS/GPS chips via native device location services (`geolocator`).
- **Continuous Position Averaging:** Enforces a 3 to 5-second continuous sampling window for single-point captures to calculate spatial mean coordinates ($\bar{x}, \bar{y}$) and counteract signal drift.
- **Accuracy Thresholds:** Displays an explicit UI warning indicator if the GPS accuracy radius exceeds 15 meters.
- **Power Optimization:** Automatically throttles GPS polling rates when accelerometer sensors detect zero movement to extend battery life during 8-hour field shifts.

FR-1.2: Camera Integration & On-Device OCR

- **Image Compression:** Automatically compresses captured images to under 1 MB per photo to optimize local device storage and reduce mobile network bandwidth during sync.
- **On-Device OCR Engine:** Integrates Google ML Kit Text Recognition to process signage text directly on the mobile device without needing an active internet connection.
- **Interactive Validation UI:** Displays extracted text suggestions beneath manual input fields for rapid selection.

FR-1.3: Offline-First Queue & Network Synchronization

- **Local Storage:** Stores all spatial vectors, metadata, and photo assets in an embedded SQLite/WatermelonDB database queue.

- **Offline Vector Tiles:** Supports pre-loaded MBTiles/Slippy map tiles for Okeho so field workers can view maps fully offline.
- **Cold-Start A-GPS Handling:** Displays a “Warming up GPS…” status indicator during off-grid cold starts to allow the device 30 to 90 seconds to lock onto orbital satellites when cell towers are unreachable.
- **Automated Sync Engine:** Batch-uploads queued local records to the central backend API automatically upon detecting active cellular data or Wi-Fi.

3.2 Backend Processing Engine & GIS Pipeline

FR-2.1: Data Schema Standardization

POINT FEATURE JSON SCHEMA:

”`json
{
“type”: “Feature”,
“geometry”: {
“type”: “Point”,
“coordinates”: [3.696120, 8.031540]
},
“properties”: {
“capture_id”: “cap_881293a”,
“feature_type”: “landmark”,
“name”: “Okeho Central Mosque”,
“ocr_suggested”: “Central Mosque”,
“gps_accuracy_m”: 3.8,
“contributor_id”: “user_41”,
“photo_path”: “https://storage.provider.com/photos/cap_881293a.jpg”,
“timestamp”: “2026-08-08T06:00:00Z”
}
}